

MedTrack_DV

Hospital Operations & Patient Analytics Dashboard

Final Project Documentation — Module 8: Documentation and Project Delivery

Project: MedTrack_DV — Hospital Operations & Patient Analytics Dashboard (Group 2)

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1. Project Overview

MedTrack_DV is an interactive Tableau dashboard suite designed to analyze hospital operations and patient analytics.

The project transforms cleaned and enriched healthcare patient-flow data into interactive dashboards covering:

- Hospital performance
- Patient admissions
- Patient flow
- Department performance
- Resource utilization

The final Tableau workbook contains four dashboards:

1. Hospital Overview
 2. Patient Flow
 3. Department Analytics
 4. Resource Utilization
-

2. Problem Statement

Healthcare organizations generate large amounts of patient and operational data.

Without an integrated analytical solution, it can be difficult to understand:

- Patient admission patterns
- Patient waiting time
- Length of stay
- Readmission
- Department performance

- Hospital occupancy
- Bed utilization
- Resource utilization

MedTrack_DV addresses this problem by providing an interactive Tableau dashboard suite for healthcare operations analysis.

The dashboard suite helps hospital administrators, healthcare managers, medical staff, and policymakers monitor patient flow, evaluate department performance, optimize hospital resources, and improve operational efficiency through data-driven decision-making.

3. Project Objectives

The project objectives are:

- Analyze hospital admissions.
- Monitor patient flow.
- Monitor occupancy.
- Analyze average length of stay.
- Analyze readmission.
- Analyze patient satisfaction.
- Compare department performance.
- Monitor bed utilization.
- Analyze healthcare resource utilization.
- Generate healthcare operational KPIs.
- Deliver an integrated Tableau dashboard suite.

4. Dataset

The Tableau workbook uses:

`cleaned_healthcare_patient_flow_data_enriched.csv`

The dataset was built from publicly available Hospital Datasets and Patient Admission Datasets, then cleaned, transformed, and enriched for healthcare operations analytics.

The dataset contains patient, admission, department, patient-flow, discharge, outcome and resource-related fields.

5. Main Dataset Fields

Patient Fields

```
patient_id
patient_age
patient_gender
patient_race
age_group
```

Admission Fields

```
patient_admission_date
patient_admission_time
admission_status
admission_shift
patient_admission_flag
admission_year_month
```

Department Fields

```
department_referral
department_group
department_efficiency_score
```

Patient Flow Fields

```
patient_waittime
waittime_category
waittime_over_45min
length_of_stay_days
```

Discharge Fields

```
patient_discharge_date
discharge_year_month
discharge_month_name
```

Outcome Fields

```
is_readmitted
readmission_status
patient_satisfaction_score
```

Resource Fields

total_beds
occupied_beds
bed_occupancy_rate

6. Healthcare KPIs

The workbook contains more than six healthcare operational KPIs:

#	KPI	Field	Aggregation
1	Total Patients	patient_id	COUNTD
2	Total Admissions	patient_admission_flag	COUNT
3	Average Length of Stay	length_of_stay_days	AVG
4	Occupancy Rate	bed_occupancy_rate	AVG
5	Readmission Rate	is_readmitted	AVG
6	Bed Utilization Rate	bed_occupancy_rate	AVG
7	Total Beds	total_beds	SUM
8	Occupied Beds	occupied_beds	SUM
9	Average Wait Time	patient_waittime	AVG
10	Patients Waiting >45 Min	waittime_over_45min	Patient count
11	Patient Satisfaction Score	patient_satisfaction_score	AVG
12	Department Efficiency Score	department_efficiency_score	AVG

Full definitions, formulas and computed values are provided in [KPI_DEFINITIONS.pdf](#).

7. Dashboards

The final Tableau workbook ([MedTrack_DV.twb](#)) contains four interconnected dashboards:

7.1 Hospital Overview

High-level summary of hospital performance and patient outcomes.

- KPI cards: Total Patients, Average LOS, Occupancy Rate, Readmission Rate, Satisfaction Score
- Charts: Admission Trends, Occupancy Rate Trend, Readmission Trend, Department Distribution, Age Group Distribution, Length of Stay

7.2 Patient Flow

Analysis of patient admissions, waiting time, admission status, admission shifts and length of stay.

- KPI cards: Total Patients, Average Wait Time, Patients Waiting >45 Min, Average LOS
- Charts: Patient Admissions Trend, Patient Admission Status, Patients by Wait Time Category, Average LOS by Department, Admission by Shift, Patient Flow Chart

7.3 Department Analytics

Comparison of operational performance across hospital departments.

- KPI cards: Admissions KPI, Average Wait Time, Average LOS, Readmission Rate, Department Efficiency Score
- Charts: Admissions by Department, Average Wait Time by Department, Readmission Rate by Department, Department Performance, Department Efficiency, Department Distribution

7.4 Resource Utilization

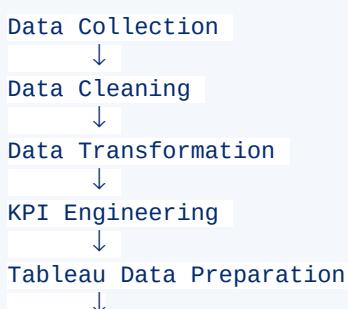
Hospital capacity, bed utilization, patient volume and department efficiency.

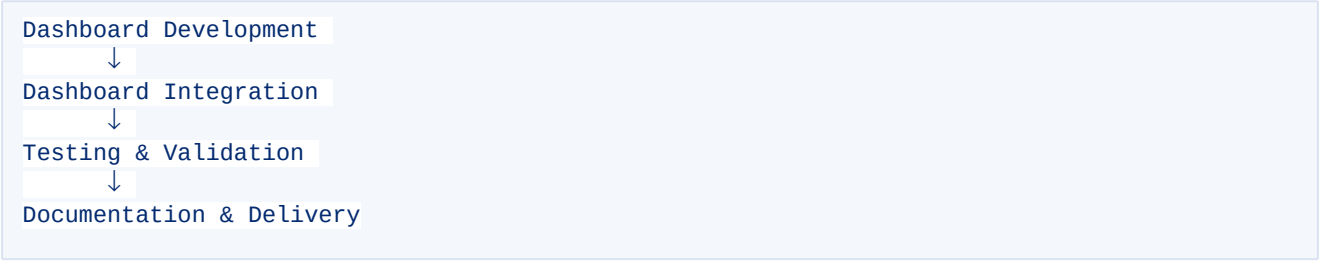
- KPI cards: Total Beds, Occupied Beds, Bed Utilization Rate, Average LOS, Average Wait Time
- Charts: Patient Volume by Department, Resource Utilization, Utilization Trend, Occupancy & Bed Utilization, Bed Occupancy by Department, Department Efficiency

Full worksheet-level detail is provided in [DASHBOARD_GUIDE.pdf](#).

8. Healthcare Operations Methodology

The project followed a nine-stage end-to-end methodology:





The complete methodology is documented in [HEALTHCARE_OPERATIONS_METHODODOLOGY.pdf](#).

9. Modules Implemented

Module	Name	Key Deliverables
Module 1	Hospital Data Collection	<code>data_collection.py</code> , integrated healthcare dataset
Module 2	Data Cleaning & Transformation	<code>data_cleaning_transformation.ipynb</code> , <code>cleaned_healthcare_patient_flow_data_en</code>
Module 3	Hospital KPI Engineering	<code>generate_hospital_kpis.py</code> , <code>hospital_final_dataset.xlsx</code>
Module 4	Dashboard Planning & Prototyping	<code>dashboard_storyboard.pdf</code>
Module 5	Hospital Overview & Patient Flow Dashboards	<code>medtrack_dashboard_v1.twb</code>
Module 6	Department Analytics & Resource Utilization Dashboards	<code>MedTrack_DV.twb</code>
Module 7	Testing & Validation	QA Checklist, Dashboard Testing Report
Module 8	Documentation & Project Delivery	Final documentation, organized repository, Tableau workbook

10. Project Structure

The repository is organized as follows:



```
■ ■■■ data_collection.py
■ ■■■ data_cleaning_transformation.ipynb
■ ■■■ generate_hospital_kpis.py
■
■■■ Module 1/ → /data – raw collected data
■ ■■■ healthcare_analytics_patient_flow_data.csv
■
■■■ Module 2/ → /data – cleaned & enriched data
■ ■■■ cleaned_healthcare_patient_flow_data_enriched.csv
■ ■■■ data_cleaning_transformation.pbix
■
■■■ Module 3/ → /data – final KPI dataset
■ ■■■ hospital_final_dataset.xlsx
■
■■■ Module 4/ → /docs – planning artifacts
■ ■■■ dashboard_storyboard.pdf
■
■■■ Module 5/ → /dashboard – development workbooks
■ ■■■ medtrack_dashboard_v1.twb
■
■■■ Module 6/ → /dashboard – development workbooks
■ ■■■ MedTrack_DV.twb
■
■■■ Dashboard/ → /dashboard – final deliverable
■ ■■■ MedTrack_DV.twb
■ ■■■ medtrack_dashboards.pbix
■
■■■ Module 7/ → /dashboard – validated final workbook
■ ■■■ MedTrack_DV.twb
■ ■■■ medtrack_dashboards.pbix
■
■■■ Module 8/ → /docs – final documentation package
■ ■■■ MODULE_8_DELIVERY_SUMMARY.md
■
■■■ Documents/ → /docs – project documentation (delivered as PDF)
■ ■■■ DATASET_SOURCES.pdf
■ ■■■ KPI_DEFINITIONS.pdf
■ ■■■ DASHBOARD_GUIDE.pdf
■ ■■■ HEALTHCARE_OPERATIONS_METHODODOLOGY.pdf
■ ■■■ FINAL_PROJECT_DOCUMENTATION.pdf
```

This maps to the required Module 8 structure (`/scripts`, `/data`, `/dashboard`, `/docs`) while preserving the module-based history of the project.

11. Tech Stack

Area	Tools / Libraries
Data Collection	Python, Hospital Dataset, Patient Admission Dataset
Data Processing	Pandas, NumPy
Data Cleaning	Python

Area	Tools / Libraries
Visualization	Tableau Desktop / Tableau Public
Dashboard Integration	Tableau Filters, Parameters, Actions
Documentation	Markdown, GitHub

12. Milestones and Timeline

Milestone	Weeks	Modules	Focus
Milestone 1	Weeks 1–2	Module 1, Module 2	Data Collection & Cleaning
Milestone 2	Weeks 3–4	Module 3, Module 4	KPI Engineering & Dashboard Planning
Milestone 3	Weeks 5–6	Module 5, Module 6	Dashboard Development & Integration
Milestone 4	Weeks 7–8	Module 7, Module 8	Testing and Delivery

13. Deployment

GitHub Repository

The complete project — scripts, datasets, dashboards and documentation — is delivered through the GitHub repository:

[Hospital-Operations-and-Patient-Analytics-Dashboard-Group-2](#)

The repository contains the organized project structure, all module deliverables and the final documentation set.

Tableau Public (Optional)

The final workbook [MedTrack_DV](#) can also be published to Tableau Public to make the dashboards viewable online without Tableau Desktop.

To publish:

1. Open [Dashboard/MedTrack_DV.twb](#) in Tableau Desktop.

2. Ensure the data source path resolves to the enriched CSV.
3. Save as `.twbx` (packaged workbook) to embed the data extract.
4. Use **Server** → **Tableau Public** → **Save to Tableau Public**.

14. Evaluation Criteria

Milestone	Focus Area	Metric	Target	Result
1	Data Collection & Cleaning	Dataset Completeness	>95% Complete	Met
2	KPI Engineering	KPI Accuracy	>95%	Met
3	Dashboard Development	Dashboard Functionality	6+ KPIs, 4 Dashboards Integrated	Met
4	Documentation & Delivery	Project Quality	Portfolio Ready	Met

Future scope and planned enhancements are described in [Section 17](#).

15. Project Outcomes

- Collection and integration of hospital datasets.
 - Data cleaning and transformation for healthcare operations analytics.
 - Development of hospital performance KPIs.
 - Creation of four interactive Tableau dashboards.
 - Patient flow and admission analysis.
 - Department efficiency monitoring.
 - Resource utilization tracking.
 - Single Tableau workbook as final deliverable.
 - Fully documented, portfolio-ready project.
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16. How to Use This Project

1. Clone or download the repository.

2. Run `Scripts/data_collection.py` to reproduce the raw integrated dataset.
3. Run `Scripts/data_cleaning_transformation.ipynb` to clean and enrich the data.
4. Run `Scripts/generate_hospital_kpis.py` to generate KPI fields and `hospital_final_dataset.xlsx`.
5. Open `Dashboard/MedTrack_DV.twb` in Tableau Desktop / Tableau Public.
6. Explore the four dashboards: Hospital Overview → Patient Flow → Department Analytics → Resource Utilization.

17. Future Scope

The MedTrack_DV platform is designed to grow beyond the delivered scope. Planned future enhancements:

1. **Real-Time Data Pipeline** — stream admissions into the dashboards live, moving from monthly batch files to hourly operational monitoring.
2. **Predictive Analytics** — forecast admissions, occupancy and length of stay with machine learning; predict readmission risk at the patient level.
3. **Alerts & Threshold Monitoring** — automatic alerts when occupancy crosses the 85% pressure threshold, waiting times exceed targets, or a department's readmission rate spikes.
4. **Bed-Level & Ward-Level Views** — drill down from hospital to ward to individual bed with real-time floor maps for capacity managers.
5. **Multi-Hospital & Regional Rollout** — benchmark departments across hospitals and regions on a shared KPI framework.
6. **Natural-Language Q&A** — let users ask the dashboard questions directly ("which department had the longest waits in June?") powered by NLP.
7. **Role-Based Access & Scheduling** — role-specific dashboard views with automated PDF/CD report distribution to management.
8. **Data Expansion** — enrich the dataset with clinical outcomes, staffing rosters and cost data to extend KPI coverage into quality and financial analytics.

18. Conclusion

MedTrack_DV demonstrates an end-to-end healthcare operations analytics workflow — from raw hospital data collection through cleaning, KPI engineering, dashboard development, integration, testing, and documentation.

The delivered Tableau workbook provides an integrated, interactive view of hospital performance, patient flow, department efficiency and resource utilization, enabling data-driven operational decision-making.

The project is fully documented and portfolio-ready.

Prepared by Group 2 under the guidance of mentor Pathina Likhita.